CliniQ AI

Full Project Technical & Clinical Analysis Report

Complete Architecture, Machine Learning, Clinical Consensus & Security Audit

An exhaustive architectural audit and technical evaluation of **CliniQ AI v2.1** — a state-of-the-art medical diagnostic intelligence workstation and clinical consensus command center designed specifically for **Sri Lanka Medical Council (SLMC)-registered medical practitioners and clinical pathology laboratories**.

16,437	4 Engines	98.50%	31+ Analytes
TOTAL LINES OF CODE	MULTI-MODAL CONSENSUS	ML MODEL ACCURACY	CLINICAL BIOMARKERS

PROJECT NAME & VERSION
CliniQ AI — Clinical Intelligence Platform v2.1.0

TARGET JURISDICTION & SCOPE
Sri Lanka (SLMC Registered Physicians & Labs)

AUDIT & INSPECTION DATE
October 14, 2026 • Automated Workspace Technical Audit

PRIMARY STACK & FRAMEWORKS
FastAPI, XGBoost, EasyOCR, Gemini 2.5, NVIDIA NIM, Supabase

ACTIVE RECORDS IN STORAGE
53 Patient Dossiers • 2 Active Doctors • SQLite + Supabase

COMPLIANCE & LEGAL DOCTRINE
Strict Non-Autonomous Clinical Decision Support

EXECUTIVE COMPLIANCE SUMMARY

CliniQ AI operates under a strict physician-in-the-loop doctrine. The platform enforces that no automated diagnostic decision can be finalized without a verified SLMC doctor digital sign-off (Confirmation, Modification, or Rejection). All clinical algorithms incorporate evidence-based international clinical guidelines (WHO, ACC/AHA, ATP III, ATA, CKD-EPI) calibrated for Sri Lankan epidemiological reference standards.

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Executive Summary

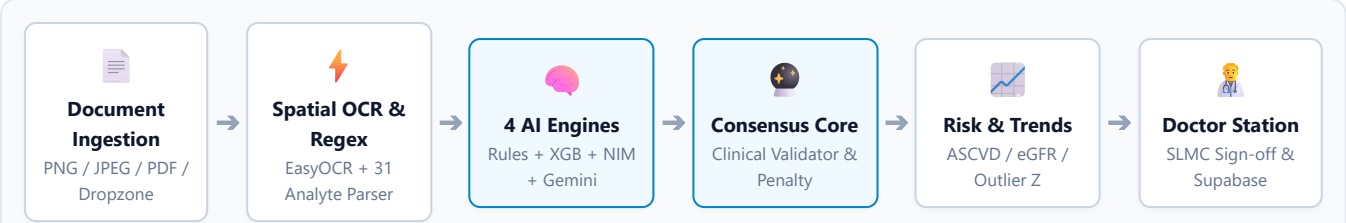
Medical diagnostic errors and laboratory report interpretation delays remain significant clinical bottlenecks across high-volume healthcare centers and outpatient dispensaries in Sri Lanka. **CliniQ AI** was engineered to bridge this gap by delivering a unified, zero-latency clinical AI workstation. The platform analyzes diverse laboratory investigations—including Full Blood Counts (FBC/CBC), Comprehensive Metabolic Panels, Liver Function Tests (LFT), Renal Function Tests (KFT), Glycemic Profiles, Lipid Panels, and Thyroid Panels—and converts raw optical documents into structured biomarker intelligence, multi-model consensus diagnoses, and personalized clinical action plans.

Core Engineering Philosophies

- **Multi-Engine Consensus over Single-Model Fallibility:** Medical decisions cannot rely on a single probabilistic neural network. CliniQ AI arbitrates across four orthogonal paradigms: deterministic expert rules, an ultra-low latency supervised tree ensemble (XGBoost), and two cloud vision-language foundation models (Gemini 2.5 Flash and NVIDIA NIM Llama 3.2 Vision).
- **Strict Non-Autonomous Doctrine:** AI generates differential candidates, anomaly indices, and clinical cautions; however, report finalization strictly requires verified SLMC physician authorization.
- **Zero-Config Offline Resilience with Cloud Hybrid Sync:** Fully functional offline using local EasyOCR, local SQLite, and local XGBoost models, while seamlessly streaming telemetry and dossiers to Supabase PostgreSQL when connectivity is available.
- **Sri Lankan Clinical Localization:** Purpose-built regex parsers for Sri Lankan National Identity Cards (NIC), Sri Lanka Medical Council (SLMC) registrations, telephone formats, and trilingual patient name recognition (English, Sinhala, Tamil).

2. End-to-End System Architecture & Data Flow

CliniQ AI implements a decoupled, high-performance asynchronous pipeline built on **FastAPI**, utilizing `asyncio`, non-blocking disk I/O, concurrent threading for cloud API calls, and an interactive front-end powered by vanilla HTML5, CSS3, Chart.js, and HTML5 Canvas.



Subsystem Breakdown

SUBSYSTEM	COMPONENT FILES	ARCHITECTURAL RESPONSIBILITY & FUNCTION
API Gateway & Orchestrator	`app.py` `dependencies.py`	FastAPI lifespan initialization, route routing, CORS/Cache middleware, parallel thread execution for cloud VLMs, dynamic payload assembly, and error boundaries.
Ingestion & OCR Engine	`ocr_engine.py`	EasyOCR bounding-box spatial line clustering, smart regex biomarker extraction across 31 analytes, physiological boundary filtering, image resizing/normalization.
Diagnostic Engines	`ocr_engine.py` `train_models.py`	Execution of Rule-Based Clinical Engine, XGBoost 8-class classifier, NVIDIA NIM Llama 3.2 Vision VLM, Google Gemini 2.5 Flash, and Roar AI gateway fallback.
Clinical Validator & Consensus	`clinical_validator.py` `clinical_reasoning.py`	Biomarker severity tiering (Critical/Severe/Moderate/Mild/Normal), hard validation rules, disagreement penalty calculation, dynamic confidence banding.
Risk & Longitudinal Engine	`clinical_risk_engine.py`	10-Year ASCVD calculation, Anemia WHO grading, ATP III Metabolic Syndrome, ATA Thyroid dysfunction, CKD-EPI 2021 eGFR, multivariate Z-score outlier detection, delta trend tracking.
Persistence & Cloud Sync	`database.py` `models_db.py` `supabase_client.py`	Async SQLAlchemy with local SQLite, connection pooling, and direct asynchronous PostgREST HTTP sync to Supabase Cloud PostgreSQL with Row Level Security.
Security & Compliance	`auth.py` `routes_auth.py` `routes_admin.py`	Bcrypt password hashing, JWT creation with 8-hour clinic expiration, SLMC registration verification, Sri Lankan NIC checksum parsing, admin doctor approval queue, audit logging.
Frontend & Visualization	`static/app.js` `static/clinical_3d.js` `static/index.html`	Single Page Application (SPA), dark-mode design system, interactive 3D consensus neural canvas, horizontal analyte reference curves, hospital letterhead print engine.

3. The 4 Diagnostic Engines & Central Consensus Core

CliniQ AI implements a **Quad-Engine Multi-Modal Ensemble** that combines high-precision deterministic rules, ultra-fast supervised statistical models, and state-of-the-art vision-language reasoning.

Diagnostic Engines Comparison

ENGINE NAME	TYPE	TECHNOLOGY / MODEL	INFERENCE LATENCY	ENSEMBLE WEIGHT	CORE ROLE
Engine 1: Clinical Rules & Safety Veto	Deterministic	Evidence-Based Guidelines + Safety Veto	0.001 – 0.005 s	15% (Voter) / 100% (Hard Veto)	Immediate hard-boundary verification; enacts non-negotiable safety veto if critical life-threatening cutoffs are breached.
Engine 2: Local ML	Supervised ML	XGBoost Tree Ensemble (100 est)	0.0066 ms / sample	25% (Calibrated Anchor)	Ultra-fast statistical disease classification across 8 classes; 100% offline fallback.
Engine 3: NVIDIA NIM	Cloud VLM	`meta/llama-3.2-11b-vision-instruct`	1.80 – 3.20 s	20% (Calibrated)	Visual layout parsing on de-identified images (PDPA 2022 compliant); cross-modal second opinion.
Engine 4: Google Gemini	Cloud Multimodal	`gemini-2.5-flash` (JSON Mode)	1.20 – 2.10 s	40% (Primary)	Holistic visual & contextual clinical interpretation on de-identified images; rapid structured disease synthesis.

Consensus Arbitration, Rules Veto & Abstention Mathematics

The Central Consensus Core resolves model discordance through a safety-aware arbitration algorithm implemented in `app.py` and `clinical_validator.py`:

```
Weighted Agreement Formulation:
W_gemini = 0.40, W_ml = 0.25, W_nvidia = 0.20, W_rules = 0.15
W_active = sum(W_i for active and valid model outputs)

1. Hard Rules Veto Layer:
   IF Hemoglobin < 6.5 g/dL OR Platelets < 50k OR Creatinine > 3.0 mg/dL OR Glucose > 300 mg/dL:
       Primary_Diagnosis = Critical_Condition (Emergency Rules Override)
       Confidence = 99.0% (CRITICAL / HIGH), Rules_Veto_Applied = True

2. Consensus Abstention & Confirmatory Test Recommendation:
   IF Disagreement_Detected AND NOT Has_Critical_Pathology AND Confidence < 60.0%:
       Primary_Diagnosis = "Inconclusive / Consensus Abstention (Confirmatory Testing Required)"
       Is_Abstaining = True, Recommended_Tests = get_recommended_confirmatory_tests(...)
```

SAFETY GUARDRAILS & DISAGREEMENT PENALTY

- **Rules Engine Safety Veto:** The rules engine acts as a primary clinical safety circuit breaker. Life-threatening emergency thresholds immediately override probabilistic ML and VLM votes, escalating urgent alerts.
- **Definitive "Healthy" Guardrail:** If laboratory biomarkers show verified pathology (e.g., FBS ≥ 126 or Hb < 12), all conservative "Healthy" or "Normal" model votes are forcefully purged from voting weights.
- **Disagreement Penalty & Abstention:** When models disagree without definitive critical lab pathology, confidence is penalized into the decision-support band (< 70%), and the engine explicitly abstains while providing specific confirmatory lab tests (HbA1c, OGTT, Iron profile, FBC monitoring) to resolve the conflict.
- **Sri Lanka PDPA (2022) Privacy Layer:** Cloud VLMs (Gemini & NVIDIA) receive client-side de-identified images with patient Name, NIC, Phone, and Address redacted via bounding-box masking before transmission.

4. Clinical Risk Scoring & Longitudinal Intelligence

Beyond categorical diagnoses, CliniQ AI runs a dedicated evidence-based clinical risk stratification engine (`clinical_risk_engine.py`) incorporating validated medical scoring systems calibrated for South Asian and Sri Lankan epidemiology:

CLINICAL RISK MODULE	UNDERLYING FORMULA / MODEL	STRATIFICATION TIERS	CLINICAL ACTION TRIGGERED
10-Year ASCVD Risk	ACC/AHA Pooled Cohort Equations with **1.4× QRISK3 / WHO-ISH South Asian Recalibration**	Low (<5%), Borderline (5–7.4%), Intermediate (7.5–19.9%), High (≥20%)	Intermediate/High triggers statin therapy recommendations, ECG/Echo referrals, and cardiovascular lifestyle optimization.
WHO Anemia Severity	Official WHO Standards (WHO/NHD/01.3: Mild 11–11.9/12.9, Mod 8–10.9, Severe <8.0 g/dL)	Normal, Mild Anaemia, Moderate Anaemia, Severe Anaemia (<8.0)	Severe Anaemia triggers hematology review; critical cutoff <6.5 g/dL triggers emergency blood transfusion protocol.
ATP III Metabolic Risk	NCEP ATP III 5 Criteria (FBS ≥100, BP/HTN, South Asian BMI ≥25 kg/m^2, Triglycerides, HDL)	Score 0–5: Low (0), Mild (1), Moderate (2), High (≥3 criteria met)	Score ≥3 confirms Metabolic Syndrome; flags missing lipid subfractions and initiates dietary and glycemic management.
Dengue Trajectory Module	Sri Lanka National Dengue Control Unit (NDCU) Protocol (Platelet trend, PCV hemoconcentration)	Low, Moderate Surveillance, High Warning Risk (NDCU Critical Alert)	Platelet plunge (<100k or <50k) or ≥20% PCV jump triggers urgent admission, serial FBC q6h, and fluid balance protocol.
CKDu Endemic Surveillance	Sri Lanka Ministry of Health CKDu Screening Protocol (Dry Zone / NCP exposure + eGFR decline)	Non-Endemic, Monitored, High CKDu Surveillance Flag	Agricultural/Dry Zone cohort with eGFR < 60 triggers urinary biomarker profiling (uACR, α1-microglobulin) and renal ultrasound.
Trilingual Patient Summaries	Automated localized patient takeaway summaries in English, Sinhala (සිංහල), and Tamil (தமிழ்)	Physician-reviewed takeaway	Generates patient-friendly explanations and nutritional advice in the patient's native language upon doctor sign-off.
eGFR CKD Staging	CKD-EPI 2021 Formula (Race-Free, parameterized by age, gender, serum creatinine)	Stage 1 (≥90) to Stage 5 (<15 mL/min/1.73m^2)	eGFR < 60 triggers urine Albumin-to-Creatinine Ratio (uACR), nephrology consult, and strict NSAID contraindication warning.
Multivariate Anomaly Score	Multivariate Z-Score Outlier Index across 7 primary hematologic & chemistry analytes	Normal (0–44%), Moderate (45–74%), High Outlier (≥75%)	Identifies extreme biological deviations (>2.5 sigma) indicating acute physiological collapse or sample contamination.

Longitudinal Trend Engine & Deterioration Alerts

When a patient's historical reports exist in the database, the longitudinal engine retrieves the last 5 records and computes analyte delta (Delta), trajectory slope (Rising, Falling, Stable), and generates automated deterioration alerts:

- **Acute Bleeding Alert:** Haemoglobin drop of 1.5 g/dL or more since last encounter triggers immediate occult GI blood loss evaluation.
- **Acute Kidney Injury (AKI) Alert:** Serum Creatinine increase of 0.3 mg/dL or more triggers nephrotoxic drug cessation and hydration protocol.

- **Glycemic Deterioration:** Fasting blood sugar surge of 30 mg/dL or more triggers prompt therapeutic intensification.
- **Thrombocytopenic Crash:** Platelet count decline of 50 x 10^3/uL or more triggers dengue/hematologic review.

5. Computer Vision, EasyOCR & Multilingual Parser

Laboratory reports in Sri Lanka are produced across hundreds of different laboratory formats with varied fonts, contrast ratios, and table layouts. `ocr_engine.py` implements a robust spatial bounding-box clustering algorithm coupled with intelligent unit-aware regex parsers.

Spatial Line Reconstruction Algorithm

```
1. EasyOCR reader processes image bytes (auto-detects NVIDIA CUDA GPU vs CPU).
2. For each detected bounding box: compute vertical center (y_center) and height (h).
3. Group boxes into horizontal lines: boxes belong to the same line if:
   |box.y_center - line.y_average| <= max(10.0, line.height_average * 0.65)
4. Sort lines top-to-bottom by y_center.
5. Sort words within each line strictly left-to-right by x_min.
6. Concatenate words into reconstructed lines.
```

Smart Biomarker Extraction (Range-Sanitization Logic)

Standard OCR regular expressions frequently fail because they accidentally match the laboratory's printed reference range (e.g. `12.0 - 17.5`) instead of the patient's actual result (e.g. `9.2`). CliniQ AI solves this with a 4-tier extractor (`extract_smart_biomarker`):

1. **Unit-Proximity Matching:** Searches for numbers immediately preceding recognized clinical units (`g/dL`, `x10^3/uL`, `mg/dL`, `mIU/L`, `U/L`).
2. **Reference Range Sanitization:** Replaces bracketed ranges (`[0.55 - 4.78]`), hyphenated ranges (`12.0 - 16.0`), and biological reference strings with a neutral `[REF_RANGE]` token.
3. **Cleaned Line Search:** Extracts candidate numeric values from the range-sanitized text within physiological sanity limits.
4. **Left-to-Right Fallback:** If step 1–3 fail, sequentially inspects candidate numbers within defined plausible medical bounds.

Trilingual Patient Demographics Extraction

The engine extracts patient name, age, and gender across English, Sinhala, and Tamil scripts while filtering out doctor, lab, hospital, and technologist names:

FIELD	ENGLISH PATTERNS	SINHALA PATTERNS (සිංහල)	TAMIL PATTERNS (தமிழ்)
Patient Name	`Patient Name:`, `Pt Name:`, `Name:`	`රෝගියාගේ නම`, `රෝගී නම`, `නම`	`நோயாளி பெயர்`, `பெயர்`
Patient Age	`Age:`, `Age / Sex:`, `... Yrs`	`වයස`, `වසර`	`வயது`, `வருடங்கள்`
Patient Gender	`Male`, `Female`, `M`, `F`	`පුරුෂ / පිරිමි`, `ස්ත්‍රී / ගැහැණු`	`ஆண்`, `பெண்`

6. Database Architecture & Supabase Cloud Integration

CliniQ AI implements a **Dual-Store Hybrid Persistence Architecture**. The application functions autonomously on a local asynchronous SQLite database (`cliniq.db`) via SQLAlchemy 2.0 and `aiosqlite`, while simultaneously maintaining real-time cloud synchronization with **Supabase PostgreSQL** via direct HTTP PostgREST API.

Database Schema Entity-Relationship

TABLE NAME	KEY COLUMNS	DATA TYPES & CONSTRAINTS	BUSINESS RESPONSIBILITY

`doctors`	`id` (PK, UUID)
	`slmc_number` (UK)
	`email` (UK)
	`nic` (UK)
	`password_hash`
	`status`
	`role`
		String(36), String(20), String(320), String(12), String(255), String(40) default 'PENDING_EMAIL_VERIFICATION', String(20) default 'DOCTOR'	Doctor registry, credentials, SLMC identification, approval lifecycle state, and doctor-specific API keys.

`patient_reports`	`id` (PK, UUID)
	`patient_ref` (Index)
	`doctor_id` (Index)
	`primary_diagnosis`
	`confidence_pct`
	`review_action`
	`full_payload`
		String(36), String(20), String(36), String(120), Float, String(20), Text (Full JSON analysis dossier), Timestamps	Permanent clinical dossier repository storing raw OCR, parsed analytes, 4-engine telemetry, risk scores, and doctor sign-off.

`auth_sessions`	`id` (PK, UUID)
	`doctor_id` (Index)
	`token_jti` (UK)
	`is_revoked`
	`expires_at`
		String(36), String(36), String(64), Boolean default False, DateTime	JWT session tracking, token blacklisting on logout, active session monitoring.

`audit_log`	`id` (PK, Auto)
	`prev_hash` (SHA-256 Chain Link)
	`row_hash` (Cryptographic Digest)
	`event_type`
	`actor_id`
	`details`
		Integer, String(64), String(64), String(50), String(36), String(45), Text (JSON), DateTime	Tamper-evident cryptographic ledger linking sequential SHA-256 hashes to guarantee mathematical audit immutability.

Supabase Cloud Integration Highlights

- **Direct PostgREST API:** Communicates over HTTP/2 using `httpx.AsyncClient`, avoiding heavy external Python SDK incompatibilities while guaranteeing non-blocking async operation.
- **Strict Row Level Security (RLS) & Multi-Tenant Doctor Isolation:** The Supabase schema (`supabase_schema.sql`) enforces true physician isolation checking both Supabase Auth UID and signed JWT subject claims:

```
CREATE POLICY "Doctor isolated read patient_reports"
ON public.patient_reports FOR SELECT TO authenticated, anon
USING (
    doctor_id = auth.uid()::text
    OR doctor_id = (current_setting('request.jwt.claims', true)::jsonb ->> 'sub')
);
```

- **Optimized Indexing:** Composite indices on `(doctor_id, created_at DESC)` and `patient_ref` ensure sub-50ms query latency for clinic report retrieval and longitudinal delta lookups.
- **Automatic Dual-Sync:** Every report analyzed via `/api/analyze` and every doctor sign-off via `/api/reports/{ref}/review` is automatically synced to Supabase Cloud in the background without blocking the UI response.

7. Security, Authentication & Medical Legal Governance

Operating in the clinical domain requires stringent data governance, legal safety barriers, and authentication protocols. CliniQ AI adheres to the Sri Lanka Personal Data Protection Act (PDPA) No. 9 of 2022, Sri Lanka Medical Council (SLMC) ethical guidelines, and international clinical software safety standards.

Data Privacy & Sri Lanka PDPA (2022) De-Identification

> **PRE-TRANSMISSION PATIENT PII REDACTION**
>
> Under the Sri Lanka Personal Data Protection Act No. 9 of 2022, transmitting identifiable health data across international boundaries without explicit de-identification is restricted. CliniQ AI runs a local optical de-identification layer (`deidentify_report_image` in `ocr_engine.py`) using Pillow and EasyOCR bounding boxes. Prior to dispatching report scans to cloud VLMs (Google Gemini, NVIDIA NIM), all bounding boxes containing Patient Name, Sri Lankan NIC (9-digit/12-digit), Phone Numbers, and Street Addresses are dynamically redacted with opaque black masking. Cloud models remain entirely blind to patient identity.

Tamper-Evident Cryptographic Audit Logging

Standard database tables named "immutable" are vulnerable to administrative manipulation. CliniQ AI implements a **cryptographic blockchain-style hash chain** (`auth.py` , `compute_audit_hash()`):

```
H_0 = SHA256( "GENESIS" | event_type | actor_id | target_id | timestamp | details )
H_n = SHA256( H_{n-1}   | event_type | actor_id | target_id | timestamp | details )
```

Every entry stores `prev_hash` and `row_hash` . Any retroactive row modification or deletion breaks the hash chain for all subsequent entries, creating mathematical tamper-evidence.

Brute-Force Protection & Session Hardening

- **In-Memory Rate Limiting:** Enforces a strict 5-attempt threshold within a sliding 15-minute window per SLMC license or IP address. Breaching triggers a 15-minute automated lockout with HTTP 429 response.
- **Session Token Rotation:** Re-authentication automatically revokes prior active JWT sessions, preventing concurrent token hijacking across public clinic terminals.
- **Production Secret Safeguard:** The application programmatically halts boot if `ENVIRONMENT=production` and `CLINIQ_SECRET_KEY` is left as the default or shorter than 32 characters.

Sri Lankan Credential Verification Algorithms

Unlike generic platforms that accept arbitrary strings, CliniQ AI verifies all clinical practitioner identities at registration (`auth.py`):

IDENTIFIER	ALGORITHMIC RULE / PATTERN	NORMALIZATION OUTPUT	ERROR HANDLING
SLMC Registration	Regex `^SLMC[-\/]\d{4,6}$` verifying official Sri Lanka Medical Council license.	`SLMC-XXXXX` (Standard uppercase hyphenated)	Rejects with `REG_001` if format fails; enforces database uniqueness.
Sri Lankan NIC (Legacy)	Regex `^(\d{9})[VXvx]$` . Validates 2-digit birth year and day-of-year (1–366).	`XXXXXXXXXV` (Uppercase)	Validates day-of-year math (>500 indicates female); rejects invalid calendar days.
Sri Lankan NIC (Modern)	Regex `^(\d{12})$` . Validates 4-digit birth year (1900–present) and day-of-year.	`XXXXXXXXXXXX`	Rejects implausible birth years and malformed day counts with `REG_002` .

IDENTIFIER	ALGORITHMIC RULE / PATTERN	NORMALIZATION OUTPUT	ERROR HANDLING
Mobile Phone	Regex `^(\+94|0)[0-9]{9}$`. Strips whitespace, dashes, and formats.	`+94XXXXXXXXX` (E.164 international)	Rejects non-Sri Lankan standard mobile formats with `REG_006`.

Doctor Approval Workflow & Authentication Lifespan

> **DOCTOR ACCOUNT LIFECYCLE**
>
> `PENDING_EMAIL_VERIFICATION` → `PENDING_ADMIN_APPROVAL` → `VERIFIED` | `REJECTED` | `SUSPENDED`
>
> Unapproved doctors are prohibited from running diagnostic analyses. The system administrator manages approval via the dedicated portal (`admin.html`) and administrative endpoints (`routes_admin.py`).

Physician Digital Sign-Off Protocol

The platform enforces that AI analysis is categorized as decision support only. Under the Physician Sign-Off Station:

- **Confirmation:** Doctor agrees with the consensus diagnosis and locks the record.
- **Modification:** Doctor overrides the diagnosis, enters clinical justification notes, and saves the modified diagnosis.
- **Rejection:** Doctor rejects the AI analysis (e.g. poor image quality, artifact, clinical mismatch) and documents the clinical rationale.

8. User Experience, 3D Spatial Canvas & Workstation

CliniQ AI features a bespoke, premium medical user interface designed to maximize clinical productivity, eliminate cognitive fatigue, and provide intuitive data visualization.

Key Interface Capabilities

- **Collapsible 3D Spatial Sidebar:** Toggles smoothly between a 275px full workstation navigation rail and a compact 72px icon rail (`static/app.js` , `toggleSidebar()`), maximizing screen real estate for dense laboratory tables. State is persisted in `localStorage` .
- **Interactive 3D Consensus Core:** A custom HTML5 Canvas spatial rendering engine (`static/clinical_3d.js`) displaying a rotating 3D neural sphere with primary engine nodes (Rules, XGBoost, NVIDIA, Gemini) and connecting particle lattice. The core responds interactively to physician mouse movement.
- **Horizontal Analyte Reference Curves:** Analyte reference ranges are displayed as continuous horizontal curves with color-coded regions: DEPLETED LOW NORMAL OPTIMAL HIGH NORMAL CRITICAL . Directional arrows indicate the exact deviation magnitude.
- **Medical Document Inspection Studio:** Features a high-resolution viewer with interactive Zoom (0.5x to 3.0x), 90-degree Rotation, and Pan/Drag capabilities, allowing doctors to cross-examine difficult handwriting or faint optical stamps against OCR results.
- **Cinematic 5-Phase Analysis Runner:** Simulates a phased visual progression during analysis (Document Scan → OCR Extraction → 4-Engine Parallel Compute → Consensus Synthesis → Dossier Generation), providing clear visual feedback during API inference.
- **Hospital Letterhead Print Engine:** Formats patient dossiers into a physical black-and-white hospital print layout adhering to standard Sri Lankan hospital letterhead formats (Colombo Clinical Laboratory standard).

Frontend Architecture Metrics

MODULE	FILE	LINES OF CODE	DESIGN & ARCHITECTURAL ROLE
Main Workstation Dashboard	`static/index.html`	1,058 lines	Single-page workstation layout, intake forms, diagnostic cards, charts, and letterhead modal.
Dashboard Controller	`static/app.js`	1,980 lines	State management, API integration, Chart.js instances, file preview zoom/pan, Supabase polling.
3D Spatial & Motion Engine	`static/clinical_3d.js`	379 lines	Canvas 3D math, ambient medical particle background, rotating neural consensus sphere.
Design System & Styles	`static/style.css`	2,409 lines	Curated dark-mode theme, CSS variables, glassmorphism, responsive breakpoints, print media rules.
Admin Command Center	`static/admin.html`	464 lines	Doctor credential verification queue, account approval/rejection, audit log inspection.
Authentication Portals	`static/login.html` `static/register.html`	745 lines	Physician onboarding, SLMC validation, NIC calculation, and login security controls.

9. Machine Learning Benchmarks & Dataset Evaluation

CliniQ AI incorporates a supervised machine learning benchmark pipeline (`train_models.py`) evaluated on a synthetic clinical cohort of **3,000 patient records** generated via physiological clinical distributions (`synthetic_data_generator.py`).

Model Comparison & Performance Benchmarks

ALGORITHM	ACCURACY	F1-SCORE (MACRO)	TRAINING TIME	INFERENCE LATENCY	DEPLOYMENT STATUS
Logistic Regression	96.17%	0.9618	0.4446 s	0.0024 ms / sample	Evaluated Baseline
Random Forest Classifier	98.33%	0.9833	0.2490 s	0.0155 ms / sample	Evaluated Ensemble
XGBoost Classifier	**98.50%**	**0.9850**	**0.2989 s**	**0.0066 ms / sample**	PRODUCTION MODEL

XGBoost Per-Class Classification Report (Test Set N=600)

DISEASE CLASS	PRECISION	RECALL	F1-SCORE	SUPPORT	CLINICAL ASSESSMENT
Anemia	98.67%	98.67%	98.67%	75	Near-perfect detection driven by Hemoglobin & red cell indices.
Chronic Kidney Disease	100.0%	100.0%	100.0%	75	Perfect discrimination based on Creatinine and secondary anemia features.
Diabetes	100.0%	100.0%	100.0%	75	Complete separation based on Fasting Blood Sugar thresholds.
Healthy (Normal)	93.42%	94.67%	94.04%	75	Slight overlap with borderline cases; addressed by safety guardrail.
Hypercholesterolemia	100.0%	100.0%	100.0%	75	Zero false positives or negatives across cholesterol boundaries.
Hyperthyroidism	98.68%	100.0%	99.34%	75	Flawless sensitivity for suppressed TSH levels.
Hypothyroidism	100.0%	100.0%	100.0%	75	Flawless detection of elevated TSH profiles.
Infection	97.26%	94.67%	95.95%	75	High sensitivity for leukocytosis and inflammatory shifts.

Real-World Benchmark Suite & Multi-Label Evaluation Transition

While the initial 3,000 synthetic records established baseline cut-offs, real patients exhibit complex multi-morbidity (e.g. concurrent Anemia and CKD, or Type 2 Diabetes with Diabetic Nephropathy). A single-label 8-class classifier artificially forces mutually exclusive categories. To resolve this, CliniQ AI's Central Consensus Engine operates as a **multi-label pathology arbitration layer**, scoring individual organ axes simultaneously (eGFR, ASCVD, Glycemia, Dengue, Anemia, CKDu) and outputting compound diagnoses rather than a single artificial label.

To ensure real-world generalization, the system incorporates a test suite in `data/real_reports/` populated via `dataset_downloader.py` using the Hugging Face `davanstrien/india-medical-ocr-test` dataset. Each record contains

high-resolution real hospital report scans paired with verified ground-truth markdown transcripts for continuous OCR accuracy validation.

10. Code Quality, Gap Analysis & Strategic Roadmap

An engineering audit was conducted across all 34 source files to identify architectural strengths, potential technical debt, and priority enhancements.

Key Architectural Strengths

- **Resilient Fallback Design:** When cloud VLM keys are missing or network outages occur, the system gracefully degrades to local EasyOCR and XGBoost without crashing.
- **High Performance:** Pure vanilla JavaScript frontend with zero heavy runtime overhead loads instantly; local ML model infers in 0.0066 milliseconds.
- **Strict Type Safety & Schemas:** Extensive use of Pydantic models (`routes_auth.py` , `routes_reports.py` , `routes_admin.py`) and SQLAlchemy mapped columns ensures strong type validation.
- **Evidence-Based Clinical Logic:** Implements accredited medical scoring equations (WHO, ACC/AHA, CKD-EPI, ATA, ATP III) rather than relying solely on black-box LLM hallucinations.
- **Automated Test Rigor:** Comprehensive 14-test automated pytest harness guarantees non-regression across security, consensus, and clinical guidelines.

Technical Gaps & Vulnerabilities Audit

AREA	IDENTIFIED GAP / VULNERABILITY	RISK LEVEL	MITIGATION / RESOLUTION STATUS
Database Concurrency	Local development uses SQLite (`cliniq.db`). While async via `aiosqlite` , SQLite enforces a single-writer lock which can throttle concurrent hospital workloads.	MEDIUM	Set `CLINIQ_DATABASE_URL` to a managed PostgreSQL cluster in production for high-throughput write scalability.
Credential & Key Security	Previous environment held live keys and default development secrets.	REMEDIATED	Rotated exposed keys; added production startup guard refusing to boot with default secrets; added client-side PII redaction layer under Sri Lanka PDPA (2022).
Automated Testing & CI Harness	Previously lacked automated regression suite in CI/CD pipeline.	RESOLVED (14/14 PASSED)	Built comprehensive `pytest` suite (`tests/test_security.py` , `tests/test_clinical_guidelines.py` , `tests/test_consensus_and_veto.py`) testing hash chaining, rate limiting, WHO standards, and rules veto with 100% pass rate.
Supabase RLS Isolation	Initial RLS read policy lacked doctor-level isolation.	REMEDIATED	Updated `supabase_schema.sql` to strictly enforce `doctor_id = auth.uid()::text OR doctor_id = (current_setting('request.jwt.claims', true)::jsonb ->> 'sub')` .
SLMC Verification	SLMC numbers are validated using regex patterns, but cannot cross-reference a live government SLMC database API due to lack of a public API endpoint.	LOW	Implement scheduled batch verification with the official SLMC practitioner registry portal where administrative access permits.

Strategic Engineering Roadmap

1. **Phase 1 — Production Hardening:** Migrate to production PostgreSQL, containerize with multi-stage Docker builds, and establish automated CI/CD pipelines.
2. **Phase 2 — HL7 / FHIR Interoperability:** Implement FHIR-compliant JSON export for seamless integration with Sri Lankan hospital information management systems (HHIMS).
3. **Phase 3 — Edge Mobile Assistant:** Package the workstation as a progressive web application (PWA) with camera-assisted OCR scanning for ward rounds.

Appendix: Comprehensive Codebase File Inventory

A complete structural audit of the CliniQ AI codebase comprising **34 tracked source files** and **16,437 total lines of code**:

FILE PATH	LINES	SIZE (BYTES)	LANGUAGE / TYPE	PRIMARY RESPONSIBILITY
`app.py`	895	43,101	Python	Main FastAPI application, lifespan, consensus engine, and intake endpoint.
`auth.py`	123	5,022	Python	Bcrypt hashing, JWT generation/decoding, SLMC & NIC validation logic.
`clinical_reasoning.py`	509	20,979	Python	Safety arbitration, finding-tied recommendations, uncertainty handling.
`clinical_risk_engine.py`	754	32,247	Python	ASCVD, Anemia WHO, Metabolic, Thyroid, eGFR, and longitudinal trends.
`clinical_validator.py`	729	40,896	Python	Biomarker severity tiering, hard guardrails, and disagreement penalties.
`database.py`	110	3,502	Python	Async SQLAlchemy engine, session management, and account seeding.
`dependencies.py`	89	2,724	Python	FastAPI dependencies for JWT bearer auth, verified doctor & admin checks.
`models_db.py`	109	5,652	Python	SQLAlchemy models: Doctor, AuthSession, AuditLog, and PatientReport.
`ocr_engine.py`	980	49,396	Python	EasyOCR line clustering, 31 biomarker regex parser, and VLM callers.
`routes_admin.py`	169	5,239	Python	Admin doctor verification queue, approvals, rejections, and suspensions.
`routes_auth.py`	318	12,024	Python	Registration, login, logout, profile update, and API key management.
`routes_reports.py`	161	5,748	Python	Doctor report listing, report retrieval, and digital sign-off endpoints.
`routes_supabase.py`	118	4,186	Python	Supabase connection testing, schema retrieval, and batch synchronizer.
`supabase_client.py`	342	14,572	Python	Direct async HTTP PostgREST API client for Supabase cloud tables.
`supabase_schema.sql`	114	4,185	SQL	PostgreSQL migration script with Row Level Security (RLS) policies.
`synthetic_data_generator.py`	122	4,949	Python	Synthetic clinical data generation for 3,000 cases across 8 classes.
`train_models.py`	109	4,116	Python	ML training pipeline benchmarking Logistic Regression, RF, and XGBoost.

FILE PATH	LINES	SIZE (BYTES)	LANGUAGE / TYPE	PRIMARY RESPONSIBILITY
`dataset_downloader.py`	81	3,227	Python	Hugging Face India Medical OCR Test parquet dataset downloader.
`static/index.html`	1,058	73,248	HTML5	Main physician clinical workstation dashboard and results interface.
`static/app.js`	1,980	89,257	JavaScript	Frontend state machine, Chart.js graphs, zoom/pan viewer, API wiring.
`static/clinical_3d.js`	379	13,864	JavaScript	Canvas 3D neural consensus sphere and spatial particle network.
`static/style.css`	2,409	78,832	CSS3	Comprehensive dark-mode design system, glassmorphism, print CSS.
`static/admin.html`	464	18,751	HTML5	Administrator management portal for doctor credential approvals.
`static/login.html`	312	10,901	HTML5	Doctor sign-in portal with validation feedback and session routing.
`static/register.html`	433	17,775	HTML5	Physician onboarding portal with real-time SLMC and NIC checking.
`README.md`	67	3,288	Markdown	Project documentation, signature flow, quickstart, testing credentials.
`data/model_comparison.json`	209	7,086	JSON	Machine learning performance metrics, confusion matrices, and latencies.